Example 5.3 Find the middle term in the expansion of $(x+y)^6$.

Solution:
Here $n = 6$; which is even. Thus the middle term in the expansion of $(x+y)^6$ is the term containing x^3y^3, that is the term ${}^6\mathrm{C}_3\, x^3y^3$ which is equal to $20x^3y^3$.

Example 5.4 Find the middle terms in the expansion of $(x+y)^7$.

Solution:
As $n = 7$ which is odd, the terms containing x^4y^3 and x^3y^4 are the two middle terms. They are ${}^7\mathrm{C}_3\, x^4y^3$ and ${}^7\mathrm{C}_4\, x^3y^4$ which are equal to $35x^4y^3$ and $35x^3y^4$.

Example 5.5 Find the coefficient of x^6 in the expansion of $(3+2x)^{10}$.

Solution:
Let us take $a = 3$ and $b = 2x$ in the binomial expansion of $(a+b)^{10}$. Then, x^6 will appear in the term containing $(2x)^6$ and nowhere else. So the term containing x^6 is

$$ {}^{10}\mathrm{C}_6\, a^4b^6 = {}^{10}\mathrm{C}_4\, a^4b^6 = \frac{10\times 9\times 8\times 7}{4\times 3\times 2\times 1}\, 3^4(2x)^6 = 210\times 3^4\times 2^6x^6. \quad [\because\ {}^{10}\mathrm{C}_6 = {}^{10}\mathrm{C}_4\,] $$

So coefficient of x^6 in the expansion of $(3+2x)^{10}$ is $210\times 3^4 2^6$.

Example 5.6 Find the coefficient of x^3 in the expansion of $(2-3x)^7$.

Solution:
Let us take $a = 2$ and $b = -3x$ in the binomial expansion of $(a+b)^7$. Then, x^3 will appear in the term containing $(-3x)^3$ and nowhere else. So the term containing x^3 is

$$ {}^7\mathrm{C}_3\, a^4b^3 = \frac{7\times 6\times 5}{3\times 2\times 1}\, 2^4(-3x)^3 = 35\times 2^4\times(-3)^3x^3. $$

So coefficient of x^3 in the expansion of $(2-3x)^7$ is $35\times 16\times(-27) = -15120$.

Example 5.7 The 2nd, 3rd and 4th terms in the binomial expansion of $(x+a)^n$ are 240, 720 and 1080 for a suitable value of x. Find x, a and n.

Solution:
It is given that $T_2 = 240$, $T_3 = 720$ and $T_4 = 1080$.

$$ \begin{aligned} T_2 &= {}^n\mathrm{C}_1\, x^{n-1}a &&= 240 \qquad (1)\\ T_3 &= {}^n\mathrm{C}_2\, x^{n-2}a^2 &&= 720 \qquad (2)\\ T_4 &= {}^n\mathrm{C}_3\, x^{n-3}a^3 &&= 1080 \qquad (3) \end{aligned} $$